Summary of How TB Learning Refined to be Neyman Orthogonal

April 3, 2026

Lemma 1 (Non-orthogonality of the density-based moment)**.** *Let $\eta_{1,k}(z,x,w) := p(A = a_k \mid z,x,w)$ denote the conditional density nuisance evaluated at anchor a_k, and define the density-based moment condition*

$$M_k^{(1)}(\gamma,\eta_1) := \mathbb{E}\Big[\eta_{1,k}(Z,X,W)\, q_\gamma(Z,a_k,X)\, \hat{h}(W,a_k,X)\Big] - \mathbb{E}\Big[\hat{h}(W,a_k,X)\Big], \tag{1}$$

where $\hat{h}(W,a_k,X)$ is the outcome bridge evaluated at anchor a_k, arising from the proximal Fredholm structure. Then $M_k^{(1)}$ is not *Neyman orthogonal with respect to $\eta_{1,k}$: for any perturbation direction $\delta\eta_1 \in L^2(P_{Z,X,W})$, the Gateaux derivative at the true values $(\gamma_0, \eta_{1,k}^0)$ satisfies*

$$\mathcal{D}_{\eta_{1,k}}\, \mathbb{E}\Big[M_k^{(1)}(\gamma_0,\eta_{1,k}^0)\Big]\,[\delta\eta_1] = \mathbb{E}\Big[\delta\eta_1(Z,X,W)\cdot q_0(Z,a_k,X)\cdot \hat{h}(W,a_k,X)\Big] \neq 0. \tag{2}$$

Consequently, estimation error in $\eta_{1,k}$ propagates at first order into the moment condition, and standard semiparametric efficiency arguments do not apply.

Proof. Differentiate $M_k^{(1)}(\gamma_0,\, \eta_{1,k}^0 + \epsilon\,\delta\eta_1)$ with respect to ϵ at $\epsilon = 0$. The result follows since $q_0 \cdot \hat{h}$ is not identically zero under the identification conditions of the proximal framework. □

Theorem 2 (Exact Neyman Orthogonality via Fredholm-Structured Correction)**.** *Let $\{a_k\}_{k=1}^K$ be anchor points covering the support of A, with anchor-specific bandwidths $\{\alpha_k\}_{k=1}^K$ where $\alpha_k > 0$. Let $K_{\alpha_k}(u) = \frac{1}{\alpha_k}K(u/\alpha_k)$ be a mollifier kernel satisfying*

$$\int K(u)\,du = 1, \qquad \int u\,K(u)\,du = 0, \qquad \kappa_2 := \int u^2\,K(u)\,du < \infty. \tag{3}$$

Define the correction term motivated by the Fredholm structure of the treatment bridge equation:

$$\phi_k^{(1)}(D;\gamma,\eta_1) := q_\gamma(Z,a_k,X)\,\hat{h}(W,a_k,X)\cdot\left[K_{\alpha_k}(A-a_k) - \eta_{1,k}(Z,X,W)\right], \tag{4}$$

and the orthogonalized score $\psi_k^{(1)} := M_k^{(1)} + \phi_k^{(1)}$, which evaluates explicitly to

$$\psi_k^{(1)}(D;\gamma) = q_\gamma(Z,a_k,X)\,\hat{h}(W,a_k,X)\,K_{\alpha_k}(A-a_k) - \hat{h}(W,a_k,X). \tag{5}$$

Then the following hold.

(i) Exact Neyman orthogonality. *For all $\delta\eta_1 \in L^2(P_{Z,X,W})$:*

$$\mathcal{D}_{\eta_{1,k}}\, \mathbb{E}\Big[\psi_k^{(1)}(D;\gamma_0)\Big]\,[\delta\eta_1] = 0. \tag{6}$$

(ii) Algebraic elimination of the density nuisance. *The nuisance $\eta_{1,k} = p(A = a_k \mid Z,X,W)$ is absent from $\psi_k^{(1)}$. Orthogonality is achieved not by debiasing in the sense of standard double*

machine learning, but by exact algebraic cancellation: the terms $\pm\,\eta_{1,k}\,q_\gamma\,\hat{h}$ cancel identically in the sum $M_k^{(1)} + \phi_k^{(1)}$, leaving only the directly observable quantity $K_{\alpha_k}(A - a_k)$. Estimation error in $\eta_{1,k}$ therefore has no effect on $\psi_k^{(1)}$, regardless of the quality of any density model.

(iii) Residual moment bias: smoothing error, not nuisance error. *At the true bridge γ_0, the tower property and a second-order Taylor expansion of $p(a_k + \alpha_k u \mid Z, X, W)$ under the kernel symmetry conditions yield:*

$$\mathbb{E}\Big[\psi_k^{(1)}(D;\gamma_0)\Big] = \underbrace{\frac{\alpha_k^2\,\kappa_2}{2}\;\mathbb{E}\Big[\partial_a^2 p(a_k \mid Z, X, W)\,q_0(Z, a_k, X)\,\hat{h}(W, a_k, X)\Big]}_{\textit{leading bias: explicit and estimable}} + O(\alpha_k^4). \tag{7}$$

The leading term has an explicit structural form determined by the curvature of the conditional treatment density at a_k. The remainder $O(\alpha_k^4)$ is the Taylor approximation error in characterizing this bias, not a second independent source of bias. Both terms depend only on the analyst-controlled bandwidth α_k and the local geometry of $p(A \mid Z, X, W)$ near a_k; neither depends on estimation error in the density nuisance.

(iv) Anchor-local bias-variance optimality. *The mean squared error of the k-th kernel-smoothed moment decomposes as*

$$\mathrm{MSE}_k(\alpha_k) = C_k^2\,\alpha_k^4 + \frac{V_k}{N\alpha_k} + O(\alpha_k^6), \tag{8}$$

where

$$C_k := \frac{\kappa_2}{2}\;\mathbb{E}\Big[\partial_a^2 p(a_k \mid Z, X, W)\,q_0(Z, a_k, X)\,\hat{h}(W, a_k, X)\Big], \tag{9}$$

$$V_k := R(K)\;\mathbb{E}\Big[q_0^2(Z, a_k, X)\,\hat{h}^2(W, a_k, X)\,p(a_k \mid Z, X, W)\Big], \tag{10}$$

with $R(K) = \int K(u)^2\,du$. The MSE is minimized by the anchor-specific bandwidth

$$\alpha_k^* = \left(\frac{V_k}{4C_k^2\,N}\right)^{1/5}, \tag{11}$$

yielding the local minimax rate $\mathrm{MSE}_k^* = O(N^{-4/5})$.

Proof sketch. Part (i). Perturb $\eta_{1,k} \to \eta_{1,k} + \epsilon\,\delta\eta_1$ in $\psi_k^{(1)}$ and differentiate at $\epsilon = 0$. This gives $\mathbb{E}[\delta\eta_1 \cdot q_0\hat{h}] - \mathbb{E}[\delta\eta_1 \cdot q_0\hat{h}] = 0$, where the two terms arise from $M_k^{(1)}$ and $\phi_k^{(1)}$ respectively.

Part (ii). Direct inspection of $M_k^{(1)} + \phi_k^{(1)}$ shows that the terms $\pm\eta_{1,k}\,q_\gamma\,\hat{h}$ cancel algebraically, yielding (5).

Part (iii). The tower property gives $\mathbb{E}[K_{\alpha_k}(A - a_k) \mid Z, X, W] = \int K_{\alpha_k}(a - a_k)\,p(a \mid Z, X, W)\,da$. Substituting $u = (a - a_k)/\alpha_k$ and Taylor expanding $p(a_k + \alpha_k u \mid Z, X, W)$ around a_k yields the stated expression; the $O(\alpha_k)$ term vanishes by kernel symmetry, and $O(\alpha_k^4)$ is the third-order Taylor remainder.

Part (iv). Minimize $C_k^2\alpha_k^4 + V_k/(N\alpha_k)$ with respect to α_k, setting the derivative to zero and solving for α_k^* as in (11). □

Remark 1 (Interpretation of the residual $O(\alpha_k^4)$ and the role of local treatment density)**.** *The $O(\alpha_k^4)$ remainder in part (iii) of Theorem 2 merits careful interpretation. It is the error in our* characterization *of the leading bias — the residual of the second-order Taylor expansion — not a*

second independent source of approximation error. Its magnitude relative to the leading $O(\alpha_k^2)$ bias term is $O(\alpha_k^2)$, which vanishes as $\alpha_k \to 0$, confirming that the leading term dominates asymptotically.

Under the optimal anchor-specific bandwidth $\alpha_k^* \propto (N\bar{p}(a_k))^{-1/5}$, where $\bar{p}(a_k) = \mathbb{E}[p(a_k \mid Z, X, W)]$ is the marginal treatment density at a_k, both the leading bias and the Taylor remainder acquire concrete N-dependent content:

$$O(\alpha_k^2) \;\propto\; O\Big((N\bar{p}(a_k))^{-2/5}\Big), \qquad O(\alpha_k^4) \;\propto\; O\Big((N\bar{p}(a_k))^{-4/5}\Big). \tag{12}$$

The remainder is thus of strictly higher order in N than the leading bias, and the approximation improves as the local effective sample size $N\bar{p}(a_k)$ grows. Three implications follow directly.

Local density and approximation quality. At anchors where $\bar{p}(a_k)$ is large, the optimal bandwidth α_k^* is small, both the leading bias and the Taylor remainder are small, and the mollifier approximation to the Dirac delta is sharp. In sparse regions α_k^* is larger, but the doubly robust product-of-errors structure — whereby the kernel smoothing bias appears multiplied by the outcome bridge residual $\hat{\varepsilon}$, giving an $O(\alpha_k^2 \cdot \delta_h)$ contribution to the final causal estimate — limits the practical impact on causal inference.

Interpretability of the leading bias. The leading bias coefficient

$$C_k = \tfrac{\kappa_2}{2}\; \mathbb{E}\Big[\partial_a^2 p(a_k \mid Z, X, W)\, q_0\, \hat{h}\Big]$$

is small precisely when the conditional treatment density is locally flat near a_k, the regime in which the bandwidth rule selects small α_k^*. The bias is therefore self-limiting in the regions that matter most for causal estimation, and the leading coefficient is identifiable from data and can serve as a diagnostic.

Higher-order kernels. For practitioners requiring tighter control of the Taylor remainder, a higher-order kernel satisfying $\int u^j K(u)\, du = 0$ for $j = 1, \ldots, s-1$ pushes the remainder to $O(\alpha_k^{2s})$, making it arbitrarily small relative to the leading bias term without altering the structure of the orthogonalized score $\psi_k^{(1)}$ or the Neyman orthogonality cancellation of part (i).

Remark 2 (Structural distinction from Wu et al. [2024])**.** *Theorem 2 differs from Wu et al. [2024] in two fundamental respects.*

The outcome bridge as test function and doubly robust attenuation. The corrected score $\psi_k^{(1)}$ uses the outcome bridge $\hat{h}(W, a_k, X)$ as the test function, a choice dictated by the proximal Fredholm structure of the treatment bridge equation. The same $\hat{h}$ appears in the doubly robust causal estimator, so the kernel smoothing bias $O(\alpha_k^2)$ in the treatment bridge is attenuated by the outcome bridge residual $\hat{\varepsilon}$ in the final estimate, giving the product-of-errors guarantee $O(\alpha_k^2 \cdot \delta_h)$. This doubly robust attenuation is a direct consequence of the Fredholm structure and is not present in Wu et al. [2024], whose moment conditions do not incorporate the outcome bridge as a test function and therefore do not admit this product form.

Anchor-specific bandwidth calibration. The bandwidth α_k is anchor-specific, calibrated to the local marginal treatment density $\bar{p}(a_k)$ via $\alpha_k^* \propto (N\bar{p}(a_k))^{-1/5}$. This ensures that the mollifier approximation to the Dirac delta is no coarser than necessary at any anchor, and that the local minimax rate $O(N^{-4/5})$ is achieved uniformly across anchors. A global bandwidth cannot achieve this: in settings where $\bar{p}(a_k)$ varies substantially across anchors — typical of nonuniform treatment distributions in observational studies — a global choice either over-smooths in dense regions or under-smooths in sparse ones. The anchor-specific rule eliminates this tradeoff by treating the bandwidth as a local parameter, consistent with both the local nature of the mollifier approximation and the local density modeling interpretation developed in Section 7.

References

Yong Wu, Yanwei Fu, Shouyan Wang, and Xinwei Sun. Doubly robust proximal causal learning for continuous treatments. *Int. Conf. Learning Representations (ICLR)*, 2024.